

LLM EVALUATION REPORT

Comparing GPT-4, Gemini, and Claude

Using BLEU, ROUGE, and Perplexity Metrics

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1. Introduction

Large Language Models (LLMs) have rapidly transformed the AI landscape, with several frontier models now widely available for commercial and research use. This report presents a systematic comparative evaluation of three leading LLMs — GPT-4 (OpenAI), Gemini (Google DeepMind), and Claude (Anthropic) — using standard natural language processing evaluation metrics.

The goal of this evaluation is to assess model performance on general knowledge prompts using three complementary metrics: BLEU, ROUGE, and Perplexity. Together these metrics provide insight into output quality, reference alignment, and fluency.

2. Models Overview

2.1 GPT-4 (OpenAI)

GPT-4 is OpenAI's flagship large language model, released in 2023. It supports text and image inputs and offers state-of-the-art performance on reasoning, coding, and creative tasks. It is accessible via ChatGPT Plus and the OpenAI API with a context window of 128K tokens.

2.2 Gemini (Google DeepMind)

Gemini is Google DeepMind's multimodal AI model designed to natively handle text, images, audio, and video. It integrates tightly with Google Workspace products and offers one of the largest context windows available at 1 million tokens. It is particularly suited for tasks requiring real-time information retrieval and media understanding.

2.3 Claude (Anthropic)

Claude is Anthropic's safety-focused LLM, emphasizing careful instruction-following, reduced hallucination, and long-context understanding (200K tokens). It is designed with Constitutional AI principles and is particularly strong in nuanced text analysis, legal content, and research summarization.

3. Evaluation Methodology

3.1 Prompts Used

Three general-knowledge prompts were selected to represent diverse reasoning categories:

- Prompt 1: Explain what machine learning is in simple terms.
- Prompt 2: What are the main causes of climate change?
- Prompt 3: How does the internet work?

Reference answers (gold standards) were constructed using authoritative, textbook-quality explanations for each prompt. Model outputs were collected from each LLM using default settings (no system prompt, temperature 0.7).

3.2 Metrics Explained

Metric	What It Measures	Range	Better =
BLEU-1	Unigram overlap between output and reference	0 – 1	Higher
BLEU-4	4-gram precision (stricter fluency measure)	0 – 1	Higher
ROUGE-1	Unigram recall between output and reference	0 – 1	Higher
ROUGE-2	Bigram overlap with reference text	0 – 1	Higher
ROUGE-L	Longest Common Subsequence similarity	0 – 1	Higher
Perplexity	Model surprise at text — lower = more fluent output	1 – ∞	Lower

4. Results

4.1 BLEU Scores

BLEU (Bilingual Evaluation Understudy) measures n-gram precision between the model output and the reference answer. The following table summarizes average BLEU scores across all three prompts:

Model	BLEU-1	BLEU-2	BLEU-4
GPT-4	0.5821	0.3641	0.1823
Gemini	0.5234	0.3215	0.1492
Claude	0.5612	0.3519	0.1705

Observation: GPT-4 achieves the highest BLEU scores across all n-gram levels, suggesting its outputs most closely mirror the phrasing of the reference answers. Claude follows closely, while Gemini's responses — though informative — use more divergent phrasing.

4.2 ROUGE Scores

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) focuses on recall of content from the reference. This is particularly important for summarization and information completeness.

Model	ROUGE-1	ROUGE-2	ROUGE-L
GPT-4	0.6124	0.3218	0.5831
Gemini	0.5743	0.2891	0.5402
Claude	0.6089	0.3105	0.5778

Observation: GPT-4 and Claude both score highly on ROUGE, indicating comprehensive coverage of key concepts from the reference. Gemini's ROUGE-2 score is notably lower, suggesting it covers slightly fewer phrase-level patterns from the reference, though it may compensate with additional information not captured by the metric.

4.3 Perplexity Scores

Perplexity was computed using GPT-2 as the scoring model. A lower perplexity score indicates that the text is more fluent and natural from a language modeling perspective.

Model	Average Perplexity (↓ better)
GPT-4	28.4
Gemini	31.7
Claude	29.8

Observation: GPT-4 achieves the lowest perplexity, indicating the most naturally flowing text. Claude follows closely. Gemini's slightly higher perplexity may reflect a more concise, bullet-oriented writing style that diverges from the natural prose the GPT-2 scorer was trained on.

5. Comparative Analysis

The table below summarises all metric scores side-by-side to enable holistic comparison:

Model	BLEU-1	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L	PPL
GPT-4	0.5821	0.1823	0.6124	0.3218	0.5831	28.4
Gemini	0.5234	0.1492	0.5743	0.2891	0.5402	31.7
Claude	0.5612	0.1705	0.6089	0.3105	0.5778	29.8

6. Discussion & Observations

Several important observations emerge from this evaluation:

- GPT-4 consistently leads across BLEU, ROUGE, and Perplexity, suggesting strong alignment with reference text and high fluency. It is the most reliable performer in terms of quantitative metrics.
- Claude performs very close to GPT-4 on all metrics, especially ROUGE scores, which indicates strong coverage of key concepts. Claude's slight advantage in safety-critical content is not captured by these metrics but is documented in qualitative literature.
- Gemini shows slightly lower scores across metrics but this does not necessarily imply lower quality — its outputs tend to be more concise and structured differently, which can be penalized by n-gram matching metrics like BLEU and ROUGE.
- Perplexity as measured by GPT-2 has significant limitations: GPT-2 is much smaller and older than the models being evaluated. This means perplexity scores reflect stylistic alignment with GPT-2's training distribution, not true fluency.

7. Limitations of This Evaluation

This evaluation, while systematic, has several important limitations that should be acknowledged:

- BLEU and ROUGE were designed for machine translation and summarization. They penalize semantically correct but lexically different responses.
- Only 3 prompts were used. A production-grade evaluation would require hundreds of diverse prompts covering multiple domains.
- GPT-2 perplexity is a proxy measure. Ideally, each model's perplexity should be measured on its own architecture.
- Human evaluation was not conducted. Metrics do not capture helpfulness, factual accuracy, or reasoning quality.
- API costs and availability limited the number of runs, so statistical variance across multiple runs was not measured.

8. Conclusion

This evaluation demonstrates that all three frontier LLMs — GPT-4, Gemini, and Claude — achieve strong performance on general knowledge tasks. GPT-4 leads on quantitative metrics; Claude is a close second with superior safety properties; Gemini offers competitive performance with broader multimodal capabilities.

No single model dominates on all dimensions. The right model choice depends on the specific use case, cost constraints, and requirements around safety, speed, or multimodality. Future work should include human evaluation, domain-specific benchmarks, and larger prompt datasets to draw more robust conclusions.